Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur
(An Autonomous Institute Affiliated to Rajasthan Technical University, Kota)



Industrial Training Presentation

TOPIC: HOSPITAL AI MANAGEMENT SYSTEM

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Roll No.:- 23ESKCX065

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Company Introduction

Celebal Technologies is a leading technology and consulting company specializing in **Artificial Intelligence, Data Engineering, Data Analytics, Cloud Computing, and Generative AI** solutions. The company focuses on developing innovative, scalable, and technology-driven solutions that help organizations improve their business processes and decision-making. During my training, I gained practical exposure to **Generative AI, Agentic AI, RAG, LLMs, database integration, and application development** while working on the **Hospital AI Management System**

CELEBAL TECHNOLOGIES

CELEBAL EXCELLENCE INTERNSHIP

CERTIFICATE

OF INTERNSHIP

This certificate is proudly awarded to

Mayuri Agrawal

Enrollment ID : **CT_CSI_DS_753**

For successfully completing the Celebal Excellence Internship 2026 at Celebal Technologies in the field of **Data Science** for the duration of **25 May 2026** to **25 Jul 2026**

Throughout the internship, you exhibited excellent problem-solving skills, dedication, and a proactive attitude toward learning and innovation. Your performance and contributions were commendable.

We appreciate your efforts and wish you continued success in your professional journey.

Sharthak Acharjee
Senior Manager



HOSPITAL AI MANAGEMENT SYSTEM

AI-Powered Multi-Agent
Hospital Management & Assistance Platform

TECHNOLOGY STACK

Python

Streamlit

Groq

LangChain

SQLite

ChromaDB

GOAL OF TRAINING



- To gain practical experience in Generative AI and Agentic AI.
- To understand how Large Language Models (LLMs) can be integrated into real-world applications.
- To learn how natural language can be converted into SQL queries.
- To implement Retrieval-Augmented Generation (RAG) for document-based question answering.
- To understand multi-agent architecture and intelligent query routing.
- To gain experience in Git, GitHub, debugging and project deployment workflow.

PROJECT OVERVIEW

PROBLEM STATEMENT

How can we create a single intelligent interface through which hospital staff can ask questions in natural language and receive relevant answers from both structured patient data and unstructured hospital documents?

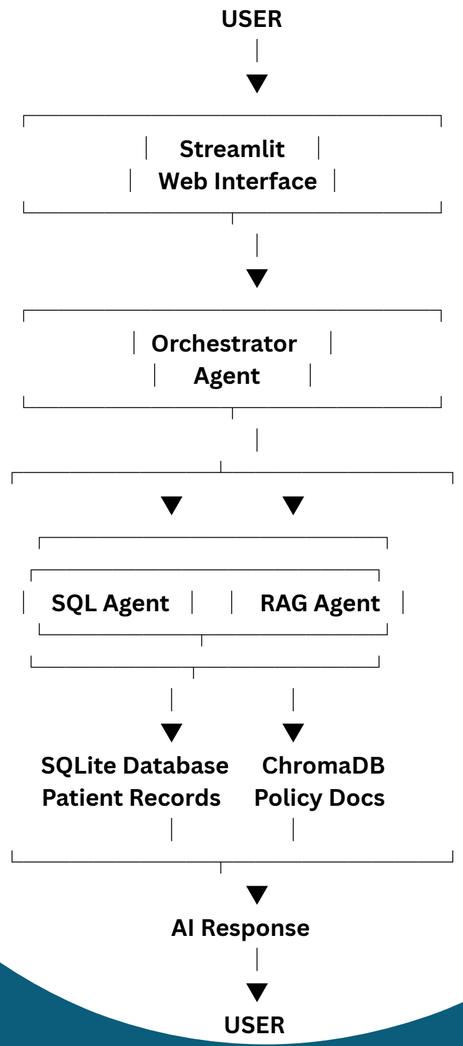
Solution

An AI-powered multi-agent Hospital Management System that intelligently determines whether a question requires:
Database Query → SQL Agent
or
Document Retrieval → RAG Agent

The Hospital AI Management System is an AI-powered hospital management and assistance platform designed to provide a unified interface for accessing hospital information.

Main Features

- 👤 Role-based login
- 🏥 Hospital dashboard
- 👨‍⚕️ Patient database
- 🤖 AI Assistant
- 📊 Analytics dashboard
- 📄 Patient reports
- 📚 Hospital policies
- ❓ Help section
- 📞 Contact section



SYSTEM ARCHITECTURE

Orchestrator Agent

decides which agent should handle the query.

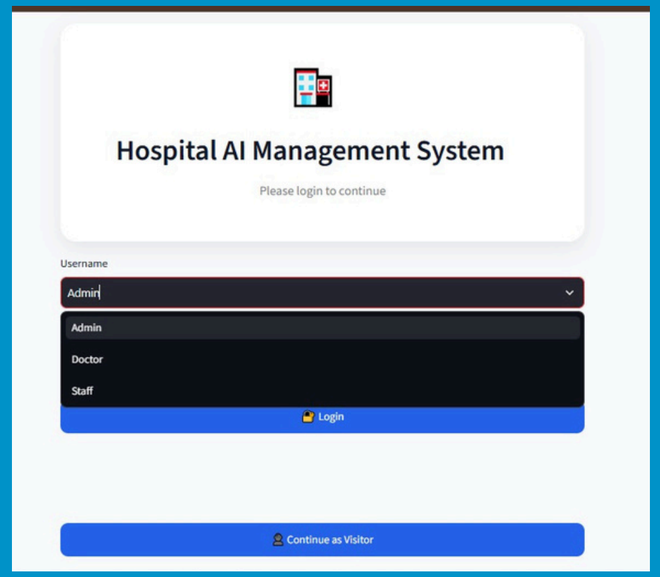
SQL Agent

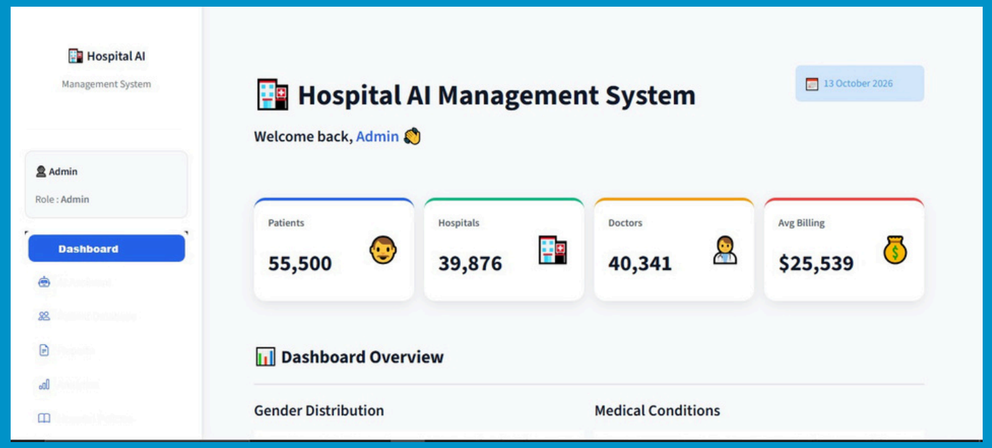
handles structured patient/database questions.

RAG Agent

handles questions related to hospital policies and documents.

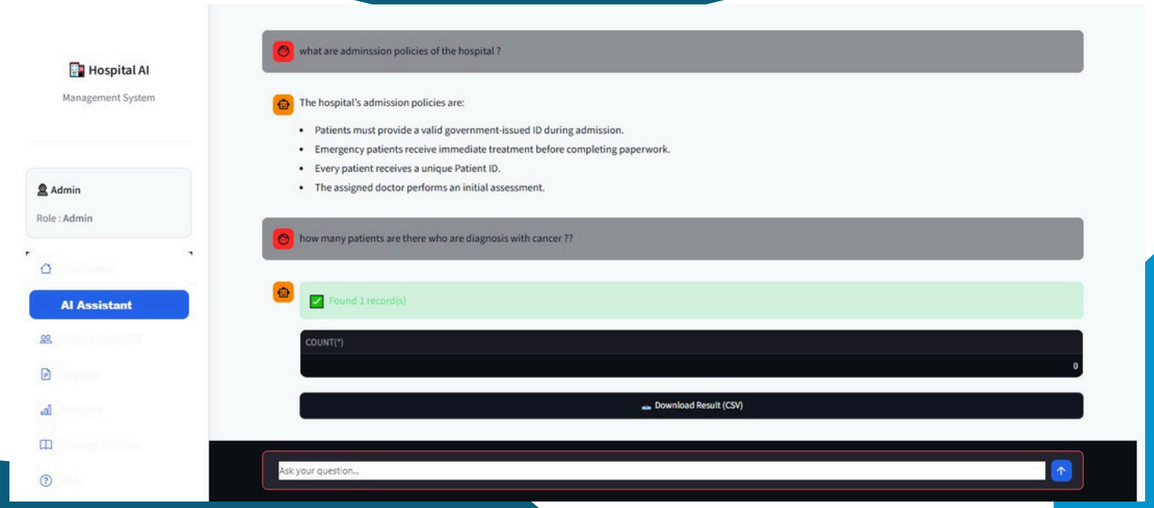
SECURE LOGIN & DASHBOARD

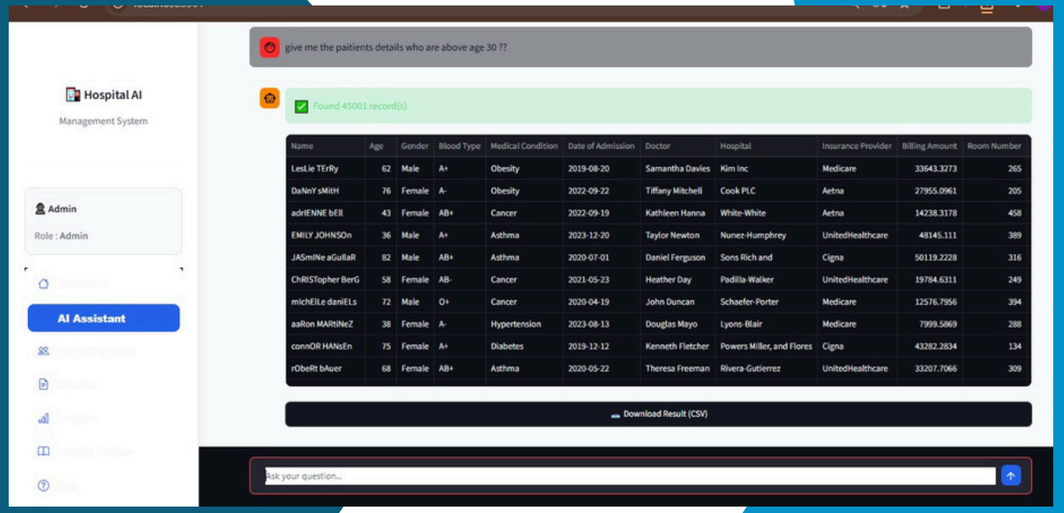




- 🔐 Role-Based Login – Admin, Doctor, Staff & Visitor
- 👤 User Identification – Displays logged-in user and role
- 📊 Dashboard Overview – Quick view of important hospital statistics
- 🏥 Hospital Insights – Patients, doctors, hospitals and billing information
- 📈 Visual Analytics – Gender distribution and medical-condition insights
- 🧭 Sidebar Navigation – Easy access to all major modules

AI ASSISTANT





The AI Assistant provides a single conversational interface through which hospital users can ask questions in natural language. An Orchestrator Agent understands the query and routes it to the appropriate specialized agent.

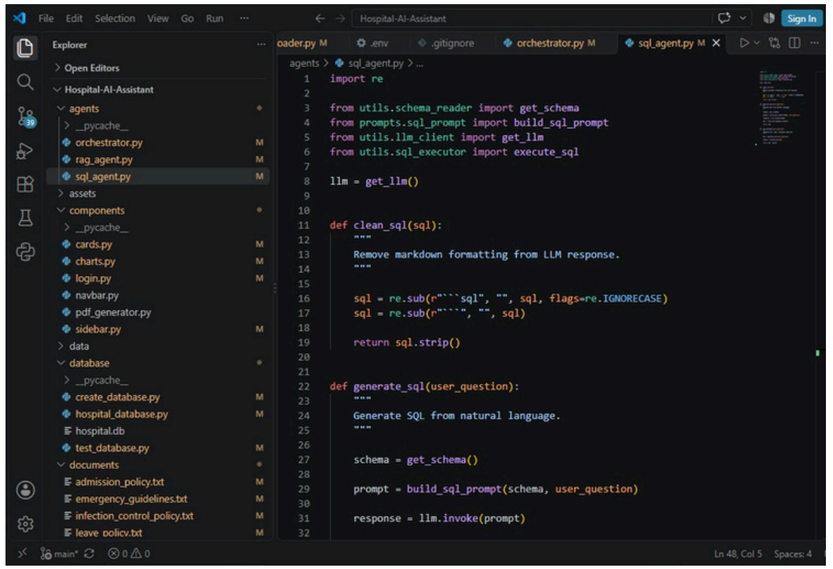
Workflow

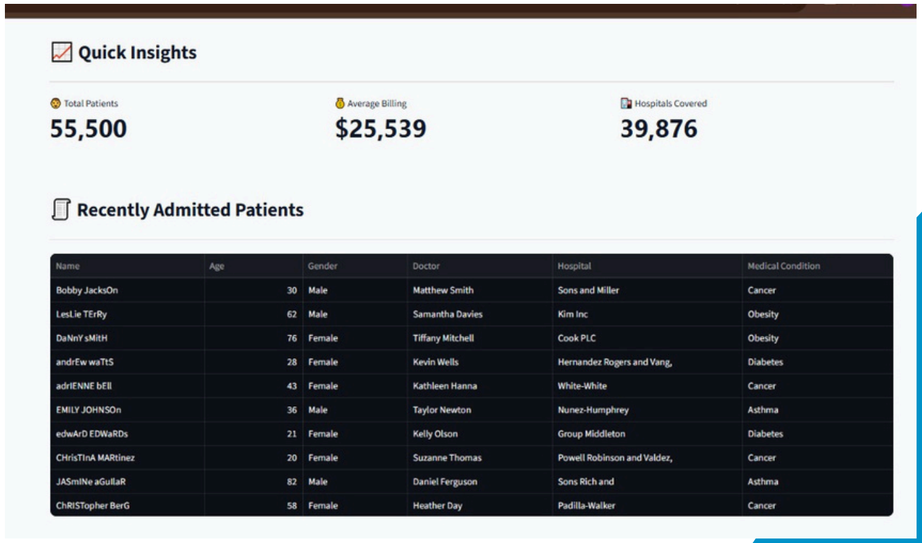
User Query
↓
Orchestrator Agent
↓
Query Classification
↓
SQL Agent / RAG Agent
↓
Information Retrieval
↓
AI-Generated Response

SQL AGENT

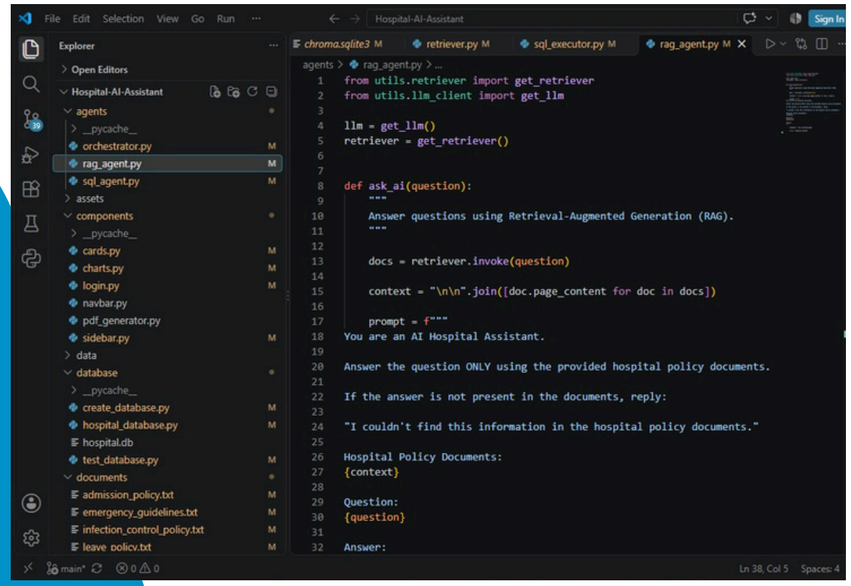
Natural Language → SQL → Answer

The SQL Agent handles questions that require information from the hospital's structured patient database. It converts natural-language questions into SQL queries, validates them and executes them against the SQLite database.



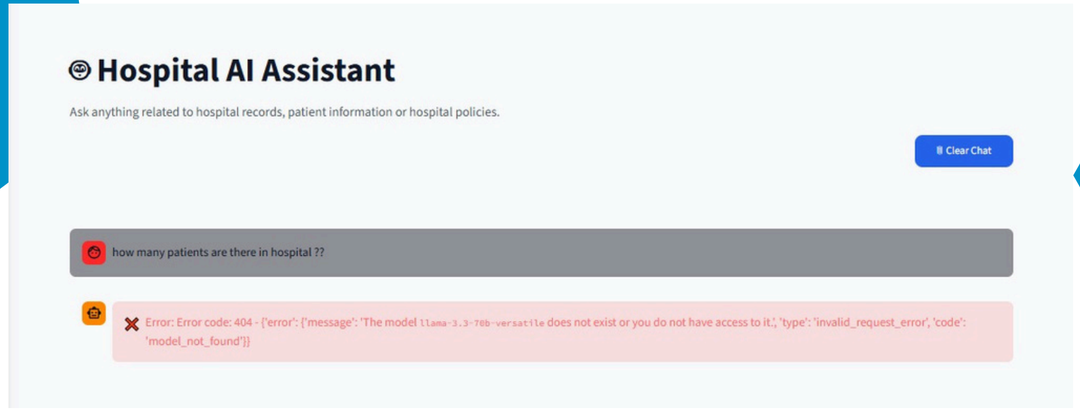


RETRIEVAL-AUGMENTED GENERATION (RAG)



The RAG Agent answers questions related to hospital policies and guidelines by retrieving relevant information from the stored policy documents before generating the final response.

Retrieves relevant hospital information first, then generates a contextual answer.



CHALLENGES & SOLUTIONS

LLM API Quota Errors
Handled API limitations and shifted the implementation toward the required Groq-based LLM workflow.

SQL Agent Errors
Fixed undefined variables, client-function mismatches and SQL-generation issues.

Database Integration
Connected the patient dataset with SQLite and implemented schema reading and query execution.

Streamlit Sidebar Issue
Investigated browser/UI behaviour and restored the custom role-based navigation sidebar.

FUTURE SCOPE

1. Cloud Deployment: Deploy the application on secure cloud infrastructure to enable remote access and improve availability across hospital departments.

2. Real Hospital Database Integration: Replace the synthetic dataset with secure, authorized hospital data to support practical use with real patient records.

3. Advanced AI Agents: Add specialized AI agents for appointment management, billing, pharmacy, doctor scheduling, and inventory management to automate hospital operations.

4. Voice-Based AI Assistant: Introduce voice commands and voice responses to help hospital staff interact with the system more conveniently.

AI

Management Systems

THANK YOU